SeizeLUT: A Self-Testing, Multiplier-Free LUT Network for Multimodal Wearable Seizure Detection on an FPGA

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***Abstract*—Wearable seizure detectors must run for days on a small battery and, being medical devices, must not fail silently. We present SeizeLUT, a seizure detector whose classifier is a Differentiable Weightless Neural Network (DWN) of 50 six-input lookup tables (LUTs) that map one-to-one onto FPGA LUTs, fed by a bit-exact, multiplier-free integer front end for behind-the-ear EEG, ECG, EMG and motion. On the official split of the SeizeIT2 dataset (125 patients, 11,626 hours), the final network reaches an event-F1 of 0.248 ± 0.016 with 0.8 false alarms per day on 29 held-out patients, above a TreeLUT model that needs 6.7 times more LUTs, a logic-gate network and the ChronoNet baseline, in 405 synthesised LUTs with no DSP or block RAM. A pre-registered two-stage variant that confirms each alarm against the post-ictal period with a second LUT network detects 73.3 % of seizures at 10.9 false alarms per hour (published SVM: 71.1 % at 11.0), and the whole cascade fits in 1,204 LUTs. Because the classifier is a set of truth tables, every single hardware fault can be enumerated: of 3,592 single faults, 2,890 change at least one clinical decision on 8,714 hours of training data, a built-in self-test fed with 1,024 consecutive recorded windows detects only 50.7 % of them, and 255 test vectors selected for coverage detect all of them; the on-chip self-test takes 10.3 µs. On a Kria KV260 the classifier runs in the fabric behind an AXI4-Lite interface with fault injection, and the integer front end runs on the Arm cores; the design is verified bit-exact on 431,960 windows at two operating points and on every injected fault.**

I. Introduction

Patients are often unaware of their seizures, so diaries undercount them [1], and wearable detection is clinically recommended today only for tonic-clonic seizures [2]. Behind-the-ear (bte) EEG is discreet but sees two channels; adding ECG, EMG and motion helps [3], [4], but the detectors that fuse them are floating-point software or multiply-accumulate networks on microcontrollers [5]. Two requirements of a wearable medical device are rarely addressed together: a classifier cheap enough for a tiny always-on fabric, and a way to know that the hardware still computes what was validated, since a silent fault turns into missed seizures or false alarms. Soft errors in FPGA configuration memory are a known risk for neural-network accelerators [6]–[8], and generic scrubbing [9] says nothing about whether a fault matters clinically.

LUT-native networks replace weighted sums by lookup tables: each neuron is a Boolean function of a few input bits that maps onto one FPGA LUT. DWNs [10], logic-gate networks [11] and LUT networks [12] train such networks by gradient descent, and tree ensembles map to LUTs in the same spirit [13]. They have been applied to ECG arrhythmia [14] and EEG classification [15], but not to seizure detection, with its extreme class imbalance (0.12 % seizure time), patient variability, missing sensors and event-level clinical scoring. Their small, explicit truth tables also make a property possible that weighted networks lack: the complete set of single hardware faults can be enumerated and judged by its clinical effect.

Contributions. (i) The first LUT-native seizure detector and the first benchmark of LUT-native models (DWN, logic-gate network, TreeLUT) against standard machine learning on the official SeizeIT2 split with SzCORE event scoring [16], with distillation from a tree ensemble to cut false alarms and modality dropout for missing sensors. (ii) A bit-exact, multiplier-free integer front end for bte-EEG, ECG, EMG and motion. (iii) A fully LUT-native post-ictal verification cascade that detects more seizures than the dataset paper's SVM at a slightly lower false-alarm rate, in 1,204 LUTs. (iv) *Self-testing*: an exhaustive single-fault analysis of the deployed classifier on 8,714 hours of data, a coverage-driven choice of on-chip test vectors that detects every decision-relevant fault with 255 vectors, and fault-injection hardware that checks this on the device. (v) A verified Kria KV260 implementation, and an independent reproduction of the published ChronoNet baseline, whose released code selects its model on the test patients.

II. Related Work

Table I compares hardware seizure detectors. ASICs reach nJ–µJ per classification with patient-specific SVMs and tree ensembles on implanted or scalp EEG [17]–[19]; an SNN on a small iCE40 FPGA detects seizures on four scalp channels [20]; transformers and fusion networks run on microcontrollers [5], [21]. Most use EEG only and epoch-level metrics; to our knowledge none reports what a hardware fault does to its decisions. On SeizeIT2, the dataset paper reports SVM and ChronoNet [22] baselines [23]; tree ensembles won the ICASSP 2023 wearable-EEG challenge [24], [25]. The only weightless seizure work is a WiSARD study on four patients [26]; for DWNs the thermometer encoder can dominate the cost [27]. Resilience studies of neural accelerators inject faults statistically [7], [8] or harden the most vulnerable layers [6]; with LUT networks the fault space is small enough to enumerate completely.

TABLE I
HARDWARE SEIZURE DETECTORS (NUMBERS AS REPORTED BY EACH WORK; PENDING VERIFICATION).

Work	Platform	Data	Signals	Evaluation	Cost
This work	KV260 FPGA	SeizeIT2, 125 pat.	bte-EEG, ECG, EMG, motion	patient-indep., held-out	405 LUTs

III. DATASET AND PROTOCOL

Data. SeizeIT2 (OpenNeuro ds005873 v1.1.0) [23]: 125 patients with focal epilepsy from five European monitoring units, 2,850 recordings, 11,626 hours of two-channel bte-EEG, single-lead ECG and EMG at 256 Hz and accelerometer and gyroscope at 25 Hz. Our audit found 883 seizures (702 in the 96 training patients, 181 in the 29 test patients). *Split and tuning.* Official split: training sub-001–096, test sub-097–125. Every choice is made by five-fold patient-wise cross-validation inside the training patients; each final model is trained on all 96 and scored once per seed on the test patients. *Scoring.* SzCORE event scoring [16] (any overlap, 30 s pre- and 60 s post-ictal tolerance, events closer than 90 s merged), averaged over patients, with false alarms per 24 h. A decision is made every second; an alarm is raised when at least k of the last 10 decisions are positive. Per-patient paired Wilcoxon tests; an extension is kept only if $p < 0.05$ in its favour.

IV. METHOD

A. Integer multimodal front end

Samples arrive as int16. Every filter is a second-order section with coefficients rounded to at most five signed powers of two and second-order error feedback. Every second, 2-s windows give per-channel and cross-channel EEG magnitude, line length, mobility, zero crossings, extrema and band magnitudes; ECG heart rate and RR variability in the log domain; EMG magnitude, line length, zero crossings and a band ratio; accelerometer and gyroscope magnitude, jerk, activity and band magnitude; presence bits; and, for most features, the difference from a causal 64-s average: 117 features without a multiplier or divider.

B. Thermometer encoding and DWN

Each of the 40 selected features (gain ranking of a LightGBM model inside each training set) is compared with 8 thresholds at training quantiles, $b_{f,i} = [x_f > t_{f,i}]$. A layer of 50 LUT6 neurons reads these bits through a learnable mapping and is trained with extended finite differences [10]; the score is the popcount of the seizure group minus that of the background group. Of seven families of extensions tested on the training folds (stateful networks, encoders, label-free adaptation, distillation, robust losses, modality handling, interpretation), two were adopted: *distillation* from a LightGBM teacher (soft targets, weight 0.5, temperature 2), and *modality dropout*, which removes ECG, EMG and motion independently with probability 0.2 per training window.

C. Post-ictal verification

Stage 1 is the final DWN at a sensitive threshold (-7, $k = 10$). Every candidate alarm is checked 96 s later by a second 50-LUT DWN on the 24 most informative of 101 statistics: 16-s ictal and 64-s post-ictal means of 45 features, their changes from the preceding 128 s, the stage-1 score and the candidate's duration. All windows are powers of two, so every mean is a sum and a shift. The settings were chosen on the training folds and pre-registered before a single test evaluation.

D. Self-testing

Fault model. The fabric classifier is 146 comparators, 50 LUT6 and an adder tree. We enumerate every single fault in the part that the network adds: each of the $50 \times 64 = 3{,}200$ truth-table bits flipped (an upset in configuration memory), and each comparator output and LUT output stuck at 0 or 1 ($292 + 100$), 3,592 faults in all. *Relevance.* Each fault is applied to all 31.4 million windows (8,714 h) of the 96 training patients. A truth-table flip at LUT j, entry a changes the score by ± 1 exactly on the windows that address (j, a), so its effect is computed in one pass; stuck-at faults are recomputed per window. A fault is *decision-relevant* if it changes at least one decision at the operating threshold, and *alarm-relevant* if it changes an alarm. *Vector selection.* A test vector detects a fault when the faulty score differs from the golden one. Since every window addresses exactly one entry per LUT, at least 64 vectors are needed when a LUT uses all its entries; we cover the addressed entries rarest first, each time taking, among the training windows that address the entry, the one that covers most uncovered entries, then add windows for stuck-at faults still undetected. *Hardware.* The built-in self-test (BIST) streams the vectors from block RAM at one per clock and compares score, decision and alarm in the fabric; fault-injection registers force any comparator or LUT output or flip any truth-table entry, so the coverage can be measured on the device.

V. RESULTS

A. Baseline reproduction

On the 29 test patients our port of ChronoNet with the released protocol detects 74.5 % of seizures at 22.6 false alarms per hour (published: 84.2 % at 100.5) and matches the published per-recording AUPR (0.109 against 0.104). The released code selects the training epoch on the test patients; a clean run (last epoch) gives 67.1 % at 11.9 false alarms per hour, and an event-F1 of 0.027 at 106 false alarms per day. The clean run is our ChronoNet reference.

TABLE II
FIVE-FOLD CROSS-VALIDATION ON THE 96 TRAINING PATIENTS (SZCORE, MEAN OVER PATIENTS; BEST-F1 OPERATING POINT; ESTIMATED HARDWARE COST).

Model	AUROC	F1	Sens	FA/24h	Cost
LightGBM	0.870	0.292	0.327	3.6	≈30.6k LUTs
CatBoost	0.863	0.286	0.337	3.0	≈27.7k LUTs
TreeLUT (5-bit leaves)	0.857	0.266	0.329	4.2	≈6.4k LUTs
DWN, 50 LUT6	0.840	0.211	0.315	2.4	≈960 LUTs
DWN + distillation (final)	0.851	0.229	0.267	0.9	≈870 LUTs
DWN + modality dropout	0.849	0.188	0.275	1.9	≈945 LUTs
Random forest	0.852	0.198	0.216	0.5	≈323k LUTs
Logic-gate network	0.843	0.171	0.330	4.9	≈7.5k LUTs
INT8 MLP	0.830	0.147	0.225	7.8	9,600 mult.

TABLE III
HELD-OUT TEST PATIENTS (29): MEAN ± STD OVER 5 SEEDS, SZCORE EVENT METRICS AT THE BEST-F1 OPERATING POINT CHOSEN ON THE TRAINING FOLDS; p: PAIRED WILCOXON ON PER-PATIENT EVENT-F1 AGAINST CLEAN CHRONONET.

Model	F1	Sens	FA/24h	p vs ChronoNet	Cost
DWN + distillation (final)	0.248±0.016	0.303	0.8	0.03	895 LUTs
DWN base	0.172±0.044	0.340	2.7	0.01	980 LUTs
DWN + modality dropout	0.242±0.030	0.272	0.4	0.03	951 LUTs
LightGBM	0.308±0.013	0.331	0.6	0.03	30,750 LUTs
XGBoost	0.294±0.014	0.301	0.4	0.06	48,295 LUTs
CatBoost	0.254±0.009	0.330	1.2	0.03	27,720 LUTs
Random forest	0.246±0.003	0.245	0.2	0.3	346,576 LUTs
Logistic regression	0.123±0.002	0.255	5.1	0.6	117 mult.
TreeLUT (5-bit leaves)	0.233±0.044	0.292	1.0	0.02	6,330 LUTs
INT8 MLP	0.155±0.008	0.249	1.7	0.02	9,600 mult.
Logic-gate network	0.191±0.021	0.291	1.4	0.03	7,452 LUTs
ChronoNet (clean)	0.027	0.790	106.3	–	CNN+GRU

B. Model comparison and missing sensors

Table II compares the model families on cross-validation. The DWN is the most compact model, with one of the lowest false-alarm rates, and beats the INT8 MLP on every metric; it remains below gradient-boosted trees ($p = 0.004$ against LightGBM). Distillation cut false alarms from 2.4 to 0.9 per day ($p = 3 \times 10^{-7}$) at a slightly higher event-F1 (0.229, $p = 0.26$) and became the final model. The base loses most of its accuracy without ECG (event-F1 0.092); modality dropout keeps it working (0.172, $p = 0.005$; without motion 0.192 against 0.136, $p = 0.021$). Out of fold, the base detects 88 % of focal to bilateral tonic-clonic seizures but 17 % of focal impaired-awareness ones.

C. Held-out test patients

The design was frozen before any of our models was scored on a test patient (Table III). The final DWN reaches an event-F1 of 0.248 ± 0.016 with 0.8 false alarms per day, above its cross-validation value: the gain from distillation holds on unseen patients (base 0.172). It is above TreeLUT (0.233) at a seventh of the LUTs, above the logic-gate network (0.191) and the INT8 MLP (0.155), level with random forest and CatBoost, and below LightGBM (0.308, $p = 0.039$). Against clean ChronoNet it is better per patient ($p = 0.027$): ChronoNet detects more seizures (79 %) but raises over a hundred times more false alarms.

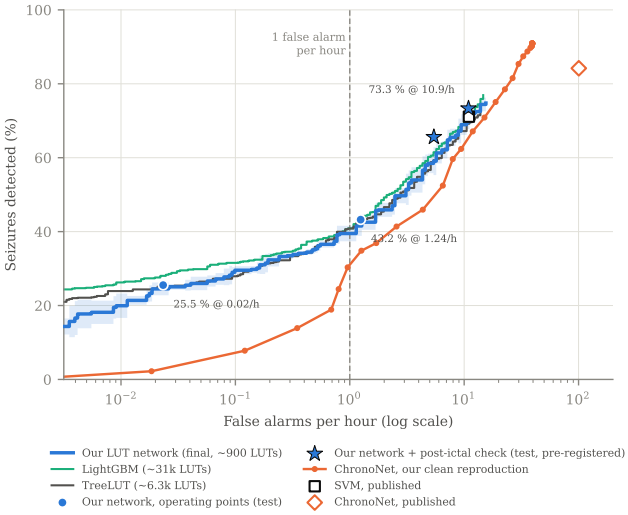


Fig. 1. Seizures detected against false alarms per hour on the 29 test patients (dataset paper's metrics). Curves: threshold sweeps of the frozen models (mean of 5 seeds); dots: operating points chosen on the training folds; stars: two-stage detector with post-ictal verification (pre-registered); open markers: published values.

D. Comparison with the dataset paper

Fig. 1 uses the dataset paper's metrics (any-overlap sensitivity and false alarms per hour). At a matched 1.2 false alarms per hour the final DWN detects 43.2 % of seizures against 34.8 % for ChronoNet, and its window AUROC is 0.883 against 0.746. The pre-registered two-stage detector reaches 73.3 ± 1.4 % at 10.9 ± 0.6 false alarms per hour over five seeds (training folds: 81.9 % at 9.6), against 71.1 % at 11.0 for the published SVM; verification adds 3.6–6.5 points over the DWN alone at the same rate. This analysis was added after the final test and is a second look at the same patients; it suits seizure logging (96-s decision delay) rather than instant alarms.

E. Hardware cost

Table IV gives synthesis results (Zynq UltraScale+, Yosys `synth_xilinx`, the same script for all, each model generated from its frozen final model and checked bit-exact against it in simulation). The DWN needs 405 LUTs, 6.7 times fewer than TreeLUT and 14.2 times fewer than the logic-gate network, with no DSP or block RAM. The post-ictal verifier, with 24-bit comparators on window sums, needs 799 LUTs; integer and floating-point verifiers agree on 99.998 % of the 118,975 training candidates. The complete two-stage detector thus costs 1,204 LUTs and no multiplier. The fault-injection and debug logic, present only in test builds, adds 298 LUTs. [board: Vivado post-implementation utilisation, timing and power of the KV260 design.]

F. Faults and self-test

Of the 3,592 single faults, 3,142 change at least one score on the training data; 450 truth-table entries (14 %) are never

TABLE IV
SYNTHESISED COST OF THE LUT-NATIVE CLASSIFIERS (YOSYS, ZYNQ ULTRASCALE+; INPUT REGISTERS, COMPARATORS, NETWORK, SCORE, DECISION AND ALARM LOGIC).

Model	LUTs	FFs	DSP	Test F1
DWN + distillation (ours)	**405**	454	0	0.248
TreeLUT (5-bit leaves)	2,721	1,455	0	0.233
Logic-gate network	5,758	2,200	0	0.191
Post-ictal verifier (stage 2)	799	637	0	–
Two-stage detector	**1,204**	1,091	0	–

TABLE V
SINGLE FAULTS OF THE FABRIC CLASSIFIER ON THE 96 TRAINING PATIENTS (8,714 H; FROZEN THRESHOLD) AND THE DECISION-RELEVANT FAULTS DETECTED (%) BY THREE BIST VECTOR SETS: 1,024 CONSECUTIVE WINDOWS OF ONE RECORDING, 1,024 RANDOM TRAINING WINDOWS AND THE 255 COVERAGE-SELECTED WINDOWS.

Fault	All	Obs.	Dec.	Alarm	Consec.	Random	**Selected**
truth-table bit flipped	3,200	2,750	2,498	1,674	43.6	79.4	**100.0**
comparator output stuck at 0/1	292	292	292	289	94.5	100.0	**100.0**
LUT output stuck at 0/1	100	100	100	100	100.0	100.0	**100.0**
all	3,592	3,142	2,890	2,063	50.7	82.2	**100.0**

addressed in 8,714 hours, so a flip there is harmless on such data. At the frozen threshold 2,890 faults are decision-relevant and 2,063 alarm-relevant (3,101 decision-relevant at the sensitive threshold). Single faults are mostly mild but not negligible: the worst truth-table flip changes 0.12 % of all decisions at the frozen threshold and the worst stuck-at fault 0.69 %; at the sensitive threshold a single stuck-at fault changes 16.3 %.

Table V and Fig. 2 show how many decision-relevant faults a BIST detects. Stuck-at faults on wires are easy: almost any data exercises them. The hard faults are flips of truth-table entries that the data rarely address, exactly the configuration-memory upsets of an FPGA. 1,024 consecutive windows of one recording around a seizure, the natural choice of a golden test, detect 50.7 % (58.9 % of the alarm-relevant ones, and 43.6 % of the decision-relevant truth-table flips): consecutive windows revisit the same entries. 1,024 random training windows reach 82.2 %. The 255 coverage-selected windows detect all observable faults, decision- and alarm-relevant at every operating point. They now open the BIST ROM, which runs 1,024 vectors in 1,028 cycles (10.3 µs at 100 MHz), so the device can test itself after every one-second window at a duty cycle of 0.001 %. In simulation, each of the 3,592 faults injected through the fault registers gives exactly the BIST error count and first failing vector that the software model predicts. [board: the same campaign on the KV260: faults injected, detected, hardware/software mismatches, time per fault.]

G. KV260 implementation

On a Kria KV260 (Zynq UltraScale+ XCK26, Fig. 3) the integer front end runs in C on the Arm Cortex-A53 cores and the classifier runs in the fabric behind an AXI4-Lite interface

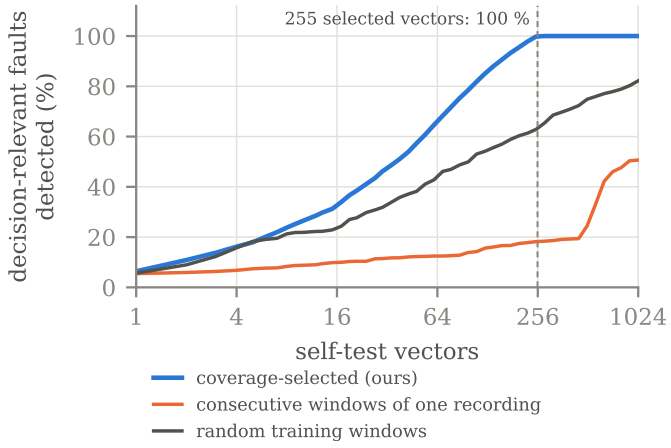


Fig. 2. Decision-relevant single faults (frozen threshold, training data) detected by the first m BIST vectors.

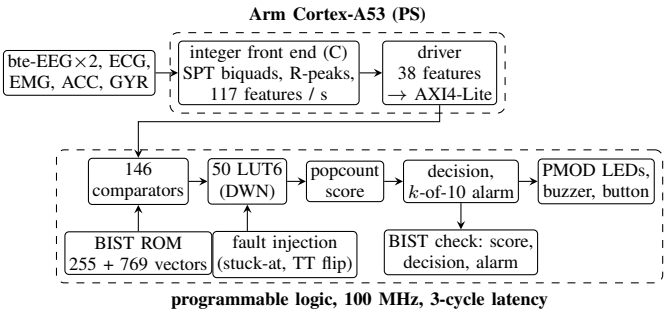


Fig. 3. SeizeLUT on the Kria KV260. The fault-injection registers and the self-test let the device check that the fabric still computes the validated classifier.

at 100 MHz, with a frozen and a programmable operating point, read-back of the 146 comparator bits and 50 LUT outputs, the BIST and fault injection. Verification before the board: the C front end, batch and second by second, reproduces all 117 features and both artefact flags of all 431,960 windows of the 10 exported test recordings exactly; the core, driven through its registers as the board software drives it, reproduces score, decision, alarm, comparator bits and LUT outputs on all 431,960 windows at two operating points (863,920 window checks, Verilator), with a latency of 3 cycles. [board: on-board replay of the 431,960 windows, measured latency and throughput, front-end time per hour of signal on the A53, SOM power from the INA260 monitor.]

VI. DISCUSSION AND LIMITATIONS

Accuracy per LUT favours the DWN by a wide margin, and the thermometer encoder, not the network, dominates its cost, as also measured by [27]; it also dominates the verifier's cost through its wide comparators. Self-testing is cheap for LUT networks because their fault space is small and their behaviour is exactly computable, which a weighted network with thousands of multipliers does not offer. The analysis covers single faults in the network part (comparator

outputs, truth tables, LUT outputs), not the adder tree, the AXI logic or the comparator constants, and relevance is measured on training data. Focal seizures remain mostly undetected, consistent with the clinical evidence [2]. Results come from retrospective recordings of one dataset.

VII. Conclusion

A 50-LUT weightless network on integer multimodal features detects seizures across patients without multipliers in 405 LUTs, reaching an event-F1 of 0.248 with 0.8 false alarms per day on 29 held-out patients, and a two-stage version in 1,204 LUTs detects more seizures than the dataset paper's SVM. Its truth tables make complete fault analysis possible: 255 selected self-test vectors detect every single fault that changes a decision, and the device can run them after every window.

Reproducibility

Code, configurations, decision log, export packages, the Verilog generator, the KV260 kit and the fault list are released with the paper. Dataset: SeizeIT2, DOI 10.18112/openneuro.ds005873.v1.1.0.

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